

Part 1: The Real Story - Why Did We Build This?

Imagine a farmer standing in a field in a small village. He notices dark spots spreading on his tomato leaves. He is worried: 'Is this a fungus? Is it a virus? Should I buy spray A or spray B?'

If he guesses wrong and buys the wrong chemical:

- He loses his money.
- The chemical burns his crops.
- His entire harvest for the year is destroyed.

Real expert crop doctors (agronomists) are very far away. So farmers want to take a photo on their mobile phone and get an instant diagnosis. But regular AI apps FAIL terribly in real farming! Here are the 3 reasons why:

1. The 'Blind Doctor' Problem (No Visual Proof)

Regular AI gives a label like 'Tomato Disease' but DOES NOT show where the disease is on the leaf! The farmer has no idea if the AI looked at the disease spot or if it just looked at background grass.

2. The 'Cocky AI' Problem (Fake 99% Confidence)

Regular AI is super overconfident. If you accidentally take a picture of your shoe, your hand, or a tractor, a regular AI will NOT say 'I don't know'. It will yell 'Tomato Late Blight - 99% Confident!' because it is forced to guess!

3. The 'Lying AI' Problem (Fake Chemical Dosages)

If you ask regular ChatGPT for chemical advice, it often hallucinates. It might invent a fake chemical name or tell the farmer to spray 10x too much chemical, which poisons the soil!

Part 2: How AgriVision AI Fixes These 3 Problems

Smart Tool 1: ResNet50 & EfficientNet AI Brains

We trained two smart AI models on 54,000 leaf photos across 38 crop types.

- * ResNet50: The accuracy champion (94.87% accurate).
- * EfficientNetV2: The mobile champion (92.20% accurate, 2x faster, uses 77% less phone memory).

Smart Tool 2: Temperature Scaling (The 'Calm Down' Filter)

We added a simple mathematical filter ($T = 1.1959$) that calms the AI down.

Instead of screaming '99% confident' on tricky photos, it calculates honest confidence scores.

Result: Reduced fake confidence errors by 61.6%!

Smart Tool 3: The Safety Rejection Guardrail

We set a strict rule: If confidence is below 60%, OR if the photo is a hand/shoe/machinery, the app STOPS and says: 'Warning: I am uncertain. Please consult a real human agronomist before spraying.'

Smart Tool 4: Grad-CAM Heatmap (The Highlighter Pen)

When the AI makes a diagnosis, it draws a glowing red/yellow heatmap over the exact disease spots on the leaf photo.

The farmer can SEE with their own eyes exactly what the AI examined!

Smart Tool 5: Knowledge-Grounded RAG (The Reference Textbook)

Instead of letting AI guess chemical remedies, we built an official 'Agri Dictionary' (rag_knowledge_base.json).

When the farmer asks for care steps, the AI reads the official dictionary FIRST, then answers.

Result: 100% verified advice with ZERO fake chemical dosages!

Part 3: How We Built This Step-by-Step (Developer Journey)

Step 1: Getting & Splitting the Leaf Data

We gathered 54,305 leaf photos. We split them carefully into 3 buckets:

- * Bucket 1 (Study Set - 70%): 37,998 photos used to teach the AI.
- * Bucket 2 (Practice Set - 15%): 8,161 photos used to adjust settings.
- * Bucket 3 (Final Exam Set - 15%): 8,146 photos kept hidden for final testing.

Step 2: Training the AI Brains

We took pre-trained AI brains that already recognized shapes (ImageNet) and trained them on leaf patterns.

- * ResNet50 scored 94.87% on the final exam.
- * EfficientNetV2 scored 92.20% and ran twice as fast.

Step 3: Creating the Agronomy Dictionary

We created rag_knowledge_base.json covering all 38 diseases. For every disease, we wrote down:

- Exact Symptoms
- Verified Chemical Spray & Dosage
- Organic / Neem Oil Remedies
- Prevention Steps

Step 4: Building the Interactive Web App

We built a clean website using Streamlit (App.py):

- Added model selection toggles (ResNet50 vs EfficientNetV2).
- Added Grad-CAM heatmap visualization.
- Added voice commands in Hindi & English.
- Added a button to download a printable PDF report card.

Part 4: How to Explain & Demo This to Anyone in 2 Minutes

Your 2-Minute Explanation Script:

"Hey! Let me show you AgriVision AI. It is an intelligent crop doctor app built for farmers.

Here is what makes it special:

- 1. It classifies 38 plant diseases with up to 94.87% accuracy using ResNet50 and EfficientNetV2.
- 2. It uses Temperature Scaling so it never gives fake 99% confidence on non-leaf photos.
- 3. It draws a glowing Grad-CAM heatmap over the leaf so you can SEE exact disease spots.
- 4. It uses RAG to look up an official agronomist reference book before answering, guaranteeing 100% accurate chemical dosages with ZERO AI hallucinations!"

Live Demo Steps:

- 1. Open Web App: Type `streamlit run App.py` in terminal.
- 2. Upload Leaf Photo: Drag and drop any plant leaf image.
- 3. View Heatmap & Diagnosis: See instant diagnosis, calibrated confidence %, and Grad-CAM heatmap.
- 4. Ask Voice Questions: Speak in Hindi or English to get care advice.
- 5. Download PDF Card: Click 'Download PDF Diagnostic Report' for a printable report.

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